MILITARY SPECIFICATION

PLASTIC SHEET, LAMINATED, METAL CLAD
(FOR PRINTED WIRING BOARDS),
GENERAL SPECIFICATION FOR

This amendment forms a part of MIL-P-13949G, dated 11 February 1987, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.2.1, delete and substitute the following:

"1.2.1 Type designation. Grade of pits and dents, class of thickness tolerances, and class of bow and twist are process parameters normally dictated by the printed-wiring manufacturing process. Unless design constraints dictate, these values shall not be included on the master drawings, but shall only be specified and used in acquisition specifications by the printed-wiring manufacturer. An asterisk shall be used to replace each digit on the master drawing."

1.2.1.1: Add a vertical dotted line to the left of Grades of pits and dents.

1.2.1.1.1: Delete "The third letter shall be as follows:" and substitute "The third letter shall represent the following special considerations:"

Add after the first sentence "If the third letter is not N or P, the second and third letter must represent the same resin system."

For definition of M, delete "opacifierf" and substitute "opacifier".

Add after last sentence in NOTE. "Also, PXP material designates paper base, epoxy resin, and flame resistant."

1.2.1.1.3, delete and substitute the following:

"1.2.1.1.3 Type and nominal weight of foil. The type and nominal weight of foil is identified by five designators as follows: The first and fourth designators shall consist of the following letters to indicate the type of foil.

A - Copper, rolled, wrought (IPC-CF-150, class 5).
B - Copper, rolled (treated).
C - Copper, drum side out, electrodeposited (IPC-CF-150, class 1).
D - Copper, drum side out (double treated) electrodeposited.
G - Copper, high ductility electrodeposited (IPC-CF-150, class 2).
H - Copper, high temperature elongation (IPC-CF-150, class 3).
J - Copper, annealed electrodeposited (IPC-CF-150, class 4).
K - Copper, light cold rolled-wrought (IPC-CF-150, class 6).
L - Copper, annealed-wrought (IPC-CF-150, class 7).
O - Unclad.
M - Copper, as rolled-wrought-low temperature (IPC-CF-150, class 8).
N - Nickel.
U - Aluminum.
Y - Copper Invar copper.

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The second and fifth designator shall include the nominal copper foil weight in ounces per square foot (oz/ft^2). The two indicators, which are separated by a slant mark third designator, shall use the actual numbers for copper foil for 1 oz/ft^2 and greater and the following letters for copper foil under 1 oz/ft^2.

E - 0.125 oz/ft^2.
Q - 0.25 oz/ft^2.
T - 0.375 oz/ft^2.
H - 0.50 oz/ft^2.
M - 0.75 oz/ft^2.
0 - Unclad.
X - For any weight or thickness not expressed (e.g., 10 oz. copperfoil) by a single digit designator."

PAGE 3

1.2.1.2, delete and substitute the following:

"1.2.1.2 Prepreg type designation.

P	GFN	7628	R	28	09	41
Prepreg material (1.2.1.2.1)	Base material (1.2.1.2.2)	Reinforcement style (1.2.1.2.3)	Flow test method (1.2.1.2.4)	Nominal resin flow (1.2.1.2.4.1)	Nominal gel time (1.2.1.2.5)	Nominal resin content (1.2.1.2.6)

P	GFN	7628	D	70	09	41
Prepreg material (1.2.1.2.1)	Base material (1.2.1.2.2)	Reinforcement style (1.2.1.2.3)	Flow test method (1.2.1.2.4)	Nominal scaled flow thickness (1.2.1.2.4.2)	Nominal gel time (1.2.1.2.5)	Nominal resin content (1.2.1.2.6)"

PAGE 4

1.2.1.2.1: Delete first sentence and substitute "The prepreg material is identified by the one-letter symbol "P", designating the construction as a preimpregnated reinforcement."

1.2.1.2.2, delete and substitute the following:

"1.2.1.2.2 Base material system. The base material is identified by three letters, with the first letter representing the reinforcement, the second letter representing the resin type and the third letter representing special considerations as shown for the laminate base material. If the third letter is not N or P, the second and third letter must represent the same resin system (see 1.2.1.1.1)."

1.2.1.2.4, delete and substitute the following:

"1.2.1.2.4 Flow test methods. Flow test method for prepreg material are identified by a letter that represents which of the two flow tests are specified. R represents nominal resin flow (see 1.2.1.2.4.1) and D represents nominal scaled flow thickness (see 1.2.1.2.4.2)."

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PAGE 6

2.3: Insert a parenthesis before "except" in the second line.

3.4.1, third line: Delete "4.8.2.2" and substitute "3.6.3".

3.4.2, delete and substitute the following:

"3.4.2 Resin systems. Resin systems used to produce laminates and prepreg under this specification shall be either epoxy, polytetrafluoroethylene, polyimide, or a combination thereof, as specified in 1.2.1.1.1 or the individual specification sheet. The composition of any laminates or prepreg represented by the manufacturer's designation or type number shall be within the following limitations:

a. The absolute amount of any component material (resin composition, curing agent, opacifier, fire resistant additive, etc.) may be varied within a total amount of ±2.5 percent of that component material.

b. The chemical species of a coloring material may be changed if the color is not substantially changed, and if the absolute amount of the coloring material is not changed by more than ±2.5 percent.

c. There shall be no additions or deletions to the composition.

The manufacturer's designation or type number shall not be included on the master drawings, but shall only be specified and used in acquisition documents by the printed wiring board manufacturer."

PAGE 7

3.4.3, delete and substitute the following:

"3.4.3 Copper foil. Copper foil shall be in accordance with IPC-CF-150. The copper foil tolerance by weight shall be class I for types C, D, G, H, and J and class II for types A, B, K, L, and M as specified in table II, unless otherwise specified (see 6.2)."

PAGE 8

TABLE II, under nominal thickness inches: Delete "0.00036" and substitute "0.0004". Also, delete "0.00052 (13.0)" and substitute "0.0005 (12.0)".

PAGE 9

TABLE III: Delete "panels" in first column.

PAGE 10

3.6.4: Delete "and 6.12" and substitute "and 6.11".

3.7.1.1.1: Delete "productibility" in NOTE and substitute "producibility".

PAGE 11

3.7.1.1.5. second paragraph, second to last line: Delete "speciments" and substitute "specimens".

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* TABLE VII. Qualification inspection.

Inspection	Requirement paragraph	Method paragraph	Specimen (form and dimension) inches	Number of specimens to be inspected	Applicable to			
					Pre-preg	Laminate		
						Type PX	0.020 inch and over	Less than 0.020 inch
Visual and dimensional inspection	3.1, 3.4, 3.5, 3.8, and 3.9	4.8.1	Complete sample sheet	1	X	X	X	X
Presence of dicyandiamide crystals	3.6.2	4.8.2.1	4xwidth	1	X	---	---	---
Thread count	3.6.3	4.8.2.2			X	---	---	---
Fabric thickness	3.6.3	4.8.2.2	4x4xthk	3	X	---	---	---
Fabric weight	3.6.3	4.8.2.2	(bias cut)		X	---	---	---
Gel time	3.6.4	4.8.2.3	Crushed B-Stage (200 ±10 mg)	3	X	---	---	---
Volatile content	3.6.5	4.8.2.4	4x4xthk (bias cut)	3	X	---	---	---
Resin content	3.6.6	4.8.2.5	4x4xthk	3	X	---	---	---
Resin flow	3.6.7	4.8.2.6.1	4x4xthk	4	X	---	---	---
Electrical strength	3.6.9	4.8.2.7.1	4x4xthk	3	X	---	---	---
Pits and dents	3.7.1.1.1	4.8.3.1	Complete sample sheet	1	---	X	X	X
Wrinkles	3.7.1.1.2				---	X	X	X
Scratches	3.7.1.1.3	4.8.3.2			---	X	X	X
Solderability	3.7.1.1.4	4.8.3.3	3x3xthk	3	---	X	X	X
Metal surfaces processability	3.7.1.1.5	4.8.3.4	12x12xthk 1/	3	---	X	X	X
Appearance of base after metal removal	3.7.1.2		12x12xthk 1/	3	---	X	X	X
Surface and subsurface imperfections	3.7.1.2.1	4.8.3.4.1	4x4xthk	3	---	X	X	X
Bow and twist (one size only)			Complete					
Sheets/panels both dimensions 12 inches or greater	3.7.2.1	4.8.3.5.1	12x12xthk	1	---	---	X	---
Panels - both dimensions under 12 inches	3.7.2.2	4.8.3.5.2	Actual size	1	---	---	X	---
Panels - one dimension under 12 inches	3.7.2.2	4.8.3.5.2	12 x smaller dimension	1	---	---	X	---
Thermal stress								
Unetched specimens	3.7.3	4.8.3.6	2x2xthk	3	---	X	X	X
Etched specimens	3.7.3	4.8.3.6	2x2xthk	3	---	X	X	X
Peel strength:		4.8.3.7		1 lengthwise				
As received	3.7.4	4.8.3.7.4	2x3xthk	for each	---	X	X	X
After thermal stress	3.7.4	4.8.3.7.1	(see fig 3)	clad side	---	X	X	X
At elevated temperature	3.7.4	4.8.3.7.2		1 crosswise	---	X	X	X
After exposure to processing solution	3.7.4	4.8.3.7.3		for each clad side	---	X	X	X
Volume resistivity and surface resistivity	3.7.5	4.8.3.8	4x4xthk or 2x2xthk	6	---	X	X	X
Dimensional stability	3.7.6	4.8.3.9	12x11xthk	3	---	X	---	X
Water absorption	3.7.7	4.8.3.10	2x2xthk	4	---	X	X	---
Dielectric breakdown (parallel to laminations)	3.7.8	4.8.3.11						
Step-by-step test	3.7.8	4.8.3.11.1	2x3xthk	3 2/	---	X	X	---
Electrical strength	3.7.9	4.8.3.12	4x4xthk	3	---	---	---	X
Permittivity and loss tangent	3.7.10	4.8.3.13	See 3/	3 3/	---	X	X	X
Flexural strength	3.7.12	4.8.3.15	See table XXVI	6 3 lengthwise 3 crosswise	---	X	X	---
Flexural strength as elevated temperature	3.7.12	4.8.3.15.1	See table XXVI	3 lengthwise	---	---	X 4/	---
Arc resistance	3.7.13	4.8.3.16	2x2xthk	3	---	X	X	X
Flammability (when applicable)	3.7.14	4.8.3.17			---	---	---	---
Prepreg	3.6.12	4.8.2.9	1x18xthk	3	X	---	---	---
Laminates 0.020 inch and greater	3.7.14	4.8.3.17	0.5x5xthk	5	---	---	X	---
Laminates less than 0.020	3.7.14	4.8.3.17	1x18xthk	3	---	X	---	X
Fungus resistance	3.7.15	4.8.3.18	2x2xthk	1	---	X	X	X
Q (resonance) (when applicable)	3.7.11	4.8.3.14	1.5x3xthk	4	---	---	X	---

1/ 12x12xthk or dimensions to provide an equivalent area such as 9x16xthk.
2/ One additional sample will be prepared for initial voltage reading for step-by-step testing.
3/ See 4.8.3.13.1, 4.8.3.13.2, 4.8.3.13.3, as applicable.
4/ GB, GH, and GI only."

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PAGE 19

TABLE VIII, delete and substitute the following:

" TABLE VIII. Base material, laminate. 1/

Qualified	Extended to
GF*	GE*
GH*	GB*
GR*	GP*
GY*	GT*, GX*
GX*	GT*
any GF*	all GF*
any GI*	all GI*

1/ * represents the special considerations (see 1.2.1.1.1)."

TABLE X: Delete title and substitute "Type of cladding.", delete "Y" (two places).

PAGE 20

TABLE XIV, delete and substitute the following:

" TABLE XIV. Base material (prepreg). 1/

Qualified	Extended to
GF*	GE*
any GF*	all GF*
any GI*	all GI*

1/ * represents the special considerations (see 1.2.1.1.1)."

TABLE XVI, delete title and substitute "Nominal resin flow and nominal scaled flow."

Add the following tables:

" TABLE XVIII-1. Class of thickness tolerance.

Qualified	Extended to
Any	All

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PAGE 34

4.8.3.17.2, delete and substitute the following:

"4.8.3.17.2 Method for laminates 0.020 inch and greater. A total of ten specimens shall be prepared for testing in accordance with IPC-TM-650, method 2.3.10."

5.3b, delete and substitute the following:

"5.3b. Manufacturer's material designation/type number and lot number."

PAGE 35

6.2c, delete and substitute the following:

"6.2c. Complete type designation (see 1.2.1), manufacturer's name, and manufacturer's type designation/type number."

PAGE 36

6.2z: Delete "and CTE".

PAGE 38

Add the following new paragraph:

"6.16 Supersession data.

Old (superseded)	New (superseding)
PC-GF**********	PGF**********
PC-GI**********	PGI**********
etc.	etc. . . . .

"

PAGE 41

FIGURE 3: Delete "3.000" and substitute "2.000".

PAGES 48 THROUGH 51

Delete, Appendix A in its entirety. This has been replaced by IPC-TM-650, method 2.5.5.3.

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PAGE 28

4.7.1.4.1, condition F: Delete "method, 263" and substitute "method 2.6.3".

PAGE 29

4.8.2.4, delete and substitute the following:

"4.8.2.4 Volatile content (see 3.6.5). Volatile content shall be measured in accordance with IPC-TM-650, method 2.3.19 except that circulating air oven shall be 225°C ±2°C for 30 minutes ±1 minute only for the base material of GIL and GIN polyimide. All others including GIJ polyimide grade shall be 163°C ±2.8°C for 15 minutes ±1 minute."

4.8.2.6.2: Delete "Scale" in two places and substitute "Scaled".

PAGE 30

4.8.2.9, delete and substitute the following:

"4.8.2.9 Flammability. A total of six specimens shall be prepared in accordance with IPC-TM-650, method 2.3.9. The manufacturer may make the prepreg into less than 0.020 inch thick laminate for this test."

4.8.3.1, delete the first sentence and substitute:

"Pits and dents shall be located visually using 20/20 vision on the as received sample sheet."

PAGE 32

4.8.3.7, delete the last sentence and substitute:

"All foil weights under 1 oz may be plated up to 1 oz but the peel strength shall be inspected using the values of the original foil weight."

4.8.3.7.3, delete and substitute the following:

"4.8.3.7.3 Peel strength after exposure to processing solutions. The peel strength shall be inspected in accordance with IPC-TM-650, method 2.4.8."

PAGE 33

4.8.3.13.1, last line: Delete "the method given in appendix A." and substitute "IPC-TM-650, method 2.5.5.3."

4.8.3.17.1, delete and substitute the following:

"4.8.3.17.1 Method for laminates under 0.020 inch. A total of six specimens shall be prepared for testing in accordance with IPC-TM-650, method 2.3.9 for flammability."

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PAGE 34

4.8.3.17.2, delete and substitute the following:

"4.8.3.17.2 Method for laminates 0.020 inch and greater. A total of ten specimens shall be prepared for testing in accordance with IPC-TM-650, method 2.3.10."

5.3b, delete and substitute the following:

"5.3b. Manufacturer's material designation/type number and lot number."

PAGE 35

6.2c, delete and substitute the following:

"6.2c. Complete type designation (see 1.2.1), manufacturer's name, and manufacturer's type designation/type number."

PAGE 36

6.2z: Delete "and CTE".

PAGE 38

Add the following new paragraph:

"6.16 Supersession data.

Old (superseded)	New (superseding)
PC-GF**********	PGF**********
PC-GI**********	PGI**********
etc.	etc. . . . .

"

PAGE 41

FIGURE 3: Delete "3.000" and substitute "2.000".

PAGES 48 THROUGH 51

Delete, Appendix A in its entirety. This has been replaced by IPC-TM-650, method 2.5.5.3.

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Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - AS, OS, SH
Air Force - 11, 85
NSA - S5
DLA - ES

User activities:
Army - AT, ME
Navy - MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
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